MIL-C-14948A (MU)
19 February 1971
SUPERSEDING
MIL-C-14948 (MU)
17 December 1968

MILITARY SPECIFICATION

CDU-10/B
LOADING, ASSEMBLING AND PACKING

1. SCOPE

1.1 This specification covers the requirements for the loading (Anti-Intrusion Devices: EAID's and SBB's), assembling and packing of one type of canister designated as CDU-10/B.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-A-2550	-	Ammunition and Special Weapons, General Specification for
MIL-R-11470	-	Radiographic Inspection, Qualification of Equipment, Operators and Procedures
MIL-I-45607	-	Inspection Equipment, Supply and Maintenance of

STANDARDS

MILITARY

MIL-STD-105	-	Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)
MIL-STD-109	-	Quality Assurance Terms and Definitions
MIL-STD-1167	-	Ammunition Data Cards

FSC: 1345

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MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

U.S. ARMY MUNITIONS COMMAND

F 9228100 - CDU-10/B
D 9229491 - XM44 Mine and Subpack Assembly
F 9234891 - Packing and Marking for Kit, Mine Canister: CDU
A 9235641 - Loading and Assembly Procedure

PUBLICATIONS

ARMY

ET 9228100 - Equipment Tabulation

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer.)

2.2 Other publications.-The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on the date of invitation for bids or request for proposal shall apply.

CODE OF FEDERAL REGULATIONS

Title 49 - Transporation, Parts 171-179

(The Department of Transportation Regulations are now a part of the Code of Federal Regulations (1949 Edition and revisions) available from the superintendent of Documents, U.S. Government Printing Office, Washington, D.C. 20402. Orders for the above publications should cite "49 CFR 171-179 (latest revision)".)

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3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with the applicable drawings and specifications.

3.2 Components

3.2.1 General.-The components shall comply with all requirements specified on applicable drawings and specifications.

3.2.2 Continuity of Liquid Level Sensor, (Dwg. 9227930) prior to assembly.-The liquid level sensor shall have electrical continuity of no more than 0.15 ohms resistance from pin E to pin H of the connector when the float is in its uppermost vertical position. The continuity and resistance (pin E to pin H) shall be continuous during the downward travel of the float for 0.062" minimum. After the float has dropped at least 0.062" from its uppermost position and before it reaches its bottommost position, the sensor shall show continuity of no more than 0.15 ohms resistance from pin E to pin D of the connector. The continuity and resistance (pin E to pin D) shall be continuous for the remaining float drop. As the float rises and before it reaches 0.062" from its uppermost position, the sensor shall again show continuity of no more than 0.15 ohms resistance from pin E to pin H. The continuity and resistance (pin E to pin H) shall be continuous as the float rises to its uppermost position. At no time shall there be simultaneous continuity of both (pin E to pin H) and (pin E to pin D). The resistance of each open circuit shall be tested and must be greater than 1 megohm.

3.2.3 Shunt verification of explosive opener fixed delay assembly (dwg. 9233359).-There shall be electrical continuity and no more than 0.3 ohms resistance measured at the termination of the detonator leads.

3.2.4 Continuity, resistance and insulation resistance of the cable assembly (dwg. 9223682) (prior to assembly).-There shall be continuity and no more than 0.1 ohm resistance between each pair of wired contacts of the cable assembly. There shall also be a minimum of 10 megohms resistance between each wire and all other wires of the P1 connector.

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3.2.5 Functioning of the electronic opener assembly (dwg. 9227463) (prior to assembly)

3.2.5.1 Minimum time delay.-The opener shall show a time delay of 1.25 seconds ± 0.25 seconds before exhibiting an output pulse. When its capacitors are charged as specified in 4.4.4.

3.2.5.2 Maximum time delay.-The opener shall show a time delay of 24.00 seconds ± 2.40 seconds before exhibiting an output pulse. When its capacitors are charged as specified in 4.4.4.

3.2.5.3 Output.-The opener shall exhibit a minimum output of one (1) volt for one hundred (100) microseconds measured across a two (2) ohm + 0.20 ohm load.

3.2.6 Continuity and resistance of the opener switch assembly (dwg. 9227439) prior to assembly

3.2.6.1 With switches in normally closed (relaxed) position.-The opener switch assembly shall show evidence of continuity and no more than 10.0 ohms resistance between the B pin of the connector and the terminal closest to the chamfered edge of the plug (blue wires) and between pin D of the connector and the terminal farthest from the chamfered edge of the plug (yellow wires). There shall also be no more than 0.5 ohm resistance between pin C of the connector and the pin receptacle (black wires).

3.2.6.2 With switches actuated.-There shall be less than 600 K ohms between pin C of the connector and the terminal closest to the chamfered edge of the plug and less than 60 K ohms between pin C of the connector assembly and the terminal farthest from the chamfered edge of the plug. A maximum force of 1000 grams shall be applied to the switch plungers to cause them to actuate.

3.2.7 Subpack (XM44 EAID) prior to loading in canisters (see dwg. 9229491)

3.2.7.1 Conditioning.-Mines contained in the subpack shall not function after immersion of the subpack in a liquid desensitizer (TCTFE) bath for twelve (12) hours minimum, and subsequent exposure in a ventilated chamber which shall be maintained at 65°F. minimum, for twelve (12) hours minimum.

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3.2.7.2 Vibration.-The mines, conditioned as specified in 3.2.7.1, shall not function when exposed to three (3) consecutive cycles of vibration applied perpendicular to the vertical axis of the subpack. Each vibration cycle shall consist of a 40 Hz, 0.20 in. double amplitude minimum for a period of thirty (30) seconds. Time between each cycle shall be thirty (30) seconds.

3.2.8 Buoyancy of XM40E5 (SBB) prior to loading in canisters (see dwg. 9225999)

3.2.8.1 Non-degassed mines.-The buoyant force for fifty (50) non-degassed mines shall be 40 grams, maximum when tested as specified in 4.4.7.

3.2.8.2 Degassed mines.-Fifty (50) degassed mines shall exhibit a buoyant force of 40 grams, maximum when tested as specified in 4.4.7.

3.3 Assembly (CDU-10/B).-The CDU-10/B shall comply with all requirements specified on drawing (dwg.) 9228100, and with all requirements specified in applicable specifications.

3.4 Desensitizer (TCTFE) leakage.-The loaded and sealed canister shall exhibit no desensitizer (TCTFE) leakage in excess of 1.0×10 cc/sec.

3.5 X-ray of loaded canister assembly

3.5.1 Desensitizer (TCTFE) level.-The desensitizer (TCTFE) level in the loaded and sealed canister when in an upright position shall be as specified on dwg. 9235641 when measured from the top of the canister. The filler and plug shall show evidence of being full of desensitizer up to the canister desensitizer level line.

3.5.2 Mine in or above filler and plug area.-The loaded and sealed canister shall not show evidence of mines above the bottom surface of the filler and plug. This includes mines positioned on top of the filler and plug or mines periphally located around the filler and plug.

3.5.3 Subpack positioning.-The subpack shall be placed in the canister with the long axis horizontal and the hinge side down. In addition the subpack shall be placed with the non-hinge end against the EOD components. Mines are permitted beneath the subpack (as a result of vibration and loading) provided that the lower surface of the subpack is no more than 1.30 inch, maximum

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above the lowest outer surface of the bottom of the canister, as specified on dwg. 9235641.

3.5.4 Filler positioning.-The filler shall be placed in the canister so that it is fully seated against the canister walls and the ends of the filler shall be tucked above the topmost side ribs of the canister. In addition, the three holes and notched surface of the filler shall be positioned toward the canister top inner surface, as specified on dwg. 9235641.

3.6 Continuity of the electrical circuits of the loaded canister assembly.-There shall be continuity between the following points:

a. Pin 4 and Pin 5 of the P1 connector.
b. Pin 7 of the P1 connector and the electronic opener mounting plate.
c. There shall be less than 600 K ohms resistance between the opener mounting plate and the terminal closest to the chamfered edge of the opener switch assembly plug.
d. There shall be less than 60 K ohms resistance between the opener mounting plate and the terminal farthest from the chamfered edge of the opener switch assembly plug.

There shall be no continuity between the following points:

e. Pin 6 of the P1 connector and the terminal closest to the chamfered edge of the opener switch assembly plug.

f. Pin 8 of the P1 connector and the terminal farthest from the chamfered edge of the opener switch assembly plug.

For purposes of this requirement, no continuity shall mean a resistance greater than 10 megohms.

3.7 Assembly (CDU Mine Canister Kit).-The CDU Mine Canister Kit shall comply with all requirements specified on dwg. 9234891 and with all requirements specified in applicable specifications.

3.8 Continuity of loaded container.-The loaded container shall show evidence of electrical continuity when tested through the monitor and ready connector (P2 connector).

3.9 Proving ground.-The loaded canister assembly shall be subjected to the following proving ground test. The following information shall be recorded for each and every test. (see 6.4)

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exposed to the atmosphere 60 seconds after being placed in a container of desensitizer (TCTFE). Test shall be performed as specified in 4.4.15.

4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to Military Standard MIL-STD-109 in order to define terms used herein. The provisions of Military Specification MIL-A-2550 shall apply.

4.1.1 Submission of product.-At the time the completed lot of product is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted:

a. A statement that the lot complies with all of the quality assurance provisions specified within this specification.
b. Drawing and specification number and date, together with an identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

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4.3 Inspection provisions

4.3.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revisions, and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to process, specification, or drawing not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168. Each inspection lot shall contain:

a. Mines of one lot interfix number from one manufacture.
b. Canister (metal parts) of one lot interfix number from one manufacturer.
c. Explosive opener fixed delay assemblies of one lot interfix number from one manufacturer.
d. Electronic openers and switch assemblies of one lot interfix number from one manufacturer.
e. Liquid level sensors of one lot interfix number from one manufacturer.
f. Fillers and plugs of one lot interfix number from one manufacturer.
g. Caps of one lot interfix number from one manufacturer.

4.3.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.3.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance

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4.3.2.2 CDU-10/B, after sealing and prior to final assembly (see dwg. 9228100)

Categories	Defects	Method of Inspection	Defect Code (See 6.2)
Critical:	None defined		
Major:	AQL 100 Percent		
101.	Incorrect torque of liquid level sensor	Test	02001
102.	Loading improperly performed	Visual	02002
Major:	AQL 1.00 Percent		
103.	Alignment of cap assembly	Gage	02003
104.	Depth of cap assembly	Gage	02004
105.	Parallelism of cap assembly	Gage	02005
106.	Excessive heat seal flashing	Visual	02006
107.	Plug not properly assembled	Visual	02007
108.	Torque of base plated mounting screws	Test	02008
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	02009

4.3.2.3 CDU-10/B (see dwg. 9228100)

Categories	Defects	Method of Inspection	Defect Code (See 6.2)
Critical:			
1.	Color of EOD marking (2 Places)	Visual	03002
Major:	AQL 100 Percent		
101.	Lanyard damaged, broken or improperly assembled	Visual	03003
102.	Plastic connectors between the opener and switch assembly not mated (prior to attaching the lid assembly and after performing the test in 4.4.10)	Visual	03004
103.	Pin receptacle not securely fastened to opener mounting plate ground pin (prior to attaching lid assembly and after performing the test in 4.4.10)	Visual/Manual	03005

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118.	Marking incorrect, misleading or unidentifiable	Visual	03020
119.	Feed-thru splice missing or loose	Visual/Manual	03021
120.	Cargo door assembled incorrectly (door plug not aligned with EOD tube)	Visual	03022
121.	Torque of lanyard retaining screw	Test	03023
122.	Improper positioning of straps and windshield pads	Visual/Manual	03024
123.	Alignment of cargo door, retention rods and retention ring relative to canister	Gage	03025
124.	Torque of cargo door screws	Gage	03026
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	03027

4.3.2.4 Container for CDU, prior to assembling crate cover (see dwg. 9234891)

Categories	Defects	Method of Inspection	Defect Code (See 6.2)
Critical:			
1.	Monitor cable connector(s) assembled to canister(s) in incorrect bay(s) (see 4.4.11)	Visual	04001
2.	Canisters improperly placed in crate	Visual	04002
3.	EOD entry holes unclear	Visual	04003
Major:	100 Percent		
101.	Ejector assembly package missing or improperly secured to platform	Visual	04004
102.	Pressure cartridge assembly package missing or improperly secured to platform	Visual	04005
Minor:	AQL 0.65 Percent		
201.	Hardware and wiring of assembled canister(s) not matching bay number(s) of canister that canister is placed in (see Table A of dwg. 9228100)	Visual	04006

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4.3.3.3 Electrical continuity, resistance and shorts of the cable assembly, prior to loading (see 3.2.4), Major Defect, Defect Code 09001.-This test shall be performed 100 percent. Any canister which fails to comply with the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.3, using equipment in accordance with 4.3.4.

4.3.3.4 Functioning of electronic opener, prior to loading (see 3.2.5), Major Defect, Defect Code 08001.-This test shall be performed 100 percent. Any opener which fails to comply with the specified requirements shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.4, using equipment in accordance with 4.3.4.

4.3.3.5 Continuity and resistance of the opener switch assembly, prior to loading (see 3.2.6), Major Defect, Defect code 08002.-A sample of 200 switches shall be selected from each lot and shall be tested as specified in 4.4.5, using equipment in accordance with 4.3.4. If any switch assembly fails to comply with the specified requirements, the lot shall be rejected.

4.3.3.6 Subpack (XM44) Conditioning and Vibration Test (see 3.2.7), Critical Defect, Defect Code 10001.-This test shall be performed 100 percent. Any subpack which fails to comply with the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.6, using equipment in accordance with 4.3.4.

4.3.3.7 Buoyant force of XM40E5 (SBB), (see 3.2.8) Major Defect, Defect Code No. 11001.-Fifty (50) assemblies shall be selected from each degassing batch or each storage container. A batch is defined as the total quantity placed in any one degassing cycle in one degassing vacuum vessel. The test shall be performed as specified in 4.4.7, using equipment specified in 4.3.4. If any sample fails to satisfy the applicable requirement, the batch or storage container shall be rejected.

4.3.3.8 (TCTFE) leakage test (see 3.4), Critical Defect, Defect Code 12001.-This test shall be performed 100 percent. Any canister which fails to comply with the specified requirements shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.8, using equipment in accordance with 4.3.4.

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4.3.3.12.2 Flight test.-A sample of ten (10) loaded canisters shall be randomly selected from each month's production and shipped to a Government-owned proving ground as designated by the Contracting Officer. The sample selected shall make up a complete reload kit (see dwg. 9234891). The test shall be conducted as specified in 4.4.12.

4.3.3.13 Tension verification of strapping machine, Major Defect, Defect Code 15001 (see 3.10).-Three (3) strap samples shall be welded and tested as specified in accordance with 4.4.13 to qualify each strapping machine for use in each production shift. In addition, one (1) sample per 160 welds shall be tested as a control of each machine during production.

If any strap sample fails during the qualification of the machine, that strapping machine shall not be used in production.

If the test sample used for control of the machine fails, three (3) additional samples shall be tested and if any sample fails, the machine shall be removed from production.

All strapping machines removed from production because of some faulty operation may, after rework, be returned to production provided they pass the above qualification test.

4.3.3.14 Leakage, Major Defect, Defect Code 16001.-This test shall be performed 100 percent. Any canister which fails to comply with the specified requirements shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.14 using equipment in accordance with 4.3.4.

4.3.3.15 Submersion, Major Defect, Defect Code 17001 (see 3.13).-This test shall be performed 100 percent. Any filler and plug assembly which fails the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.15, using equipment in accordance with 4.3.4.

4.3.4 Inspection equipment.-Equipment Tabulation (ET) number 9228100 and Equipment Lists (EL's) referenced thereon identify the inspection equipment required to perform the examinations and tests prescribed in this section. The contractor shall design inspection equipment in accordance with the instructions in 6.3.

4.3.4.1 Government rights to documentation.-Inspection equipment drawings and lists provided and revised in accordance with the requirements of this specification may be used by DOD activities for design, procurement, manufacture, testing,

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4.4.6 Subpack (XM44 EAID) conditioning and vibration, prior to loading in canisters (see 3.2.7).-Completely immerse subpack in desensitizer for a minimum of 12 hours. Remove subpacks from desensitizer and place in a ventilated area at a temperature of 65°F or greater for 12 hours minimum to arm mines. Secure subpacks to a slip table or other surface capable of imparting a motion to the subpacks as described below. Mount subpack with long axis vertical, the small end down and the hinged end up. The motion must be in a plane perpendicular to the long axis of the subpack. The subpacks must be sucurely mounted to holding fixtures and separated to permit free movement of air to at least five of the six sides of the subpack. Racks must be securely mounted to holding fixtures and separated to permit free movement of air to at least five of the six sides of the subpack. Racks may be used to vibrate a large number of subpacks at one time, but separation of each tier of subpacks must be maintained for free air movement. After completion of the vibration cycles, observation shall be made for possible functioning of one or more of the XM44"s in each subpack. Functioning of an XM44 is evidenced by the presence of smoke in the vicinity of the functioned unit. The fact that a unit has functioned will often be indicated by the odor of burned powder in the rea, the sound of the detonation, or a flash. Detection of these, however, depend on the ventilation, the ambient noise level during the test or viewing facilities in the test chamber. Any subpack in which one or more XM44 functions, will be separated from the rest and palced in freon for at least 12 hours until it can be removed to a disposal area for destruction. Subpacks showing no evidence of XM44's functioning will be immediately placed in desensitizer and held for at least 12 hours before being moved. They may then be moved to a storage or holding area and kept immersed in desensitizer until being placed in CDU-10/B canisters.

CAUTION: Exercise adequate safety precautions during movement and handling of subpacks after exposure to conditioning and vibration.

4.4.7 Buoyancy of XM40E5 (SBB) prior to loading (see 3.2.8).-Any approved, non-buoyant container shall be submerged in desensitizer and attached to a beam balance. The beam balance shall now be balanced with the container submerged and not in contact with any supporting surface except the beam balance. Record the beam balance reading (tare weight). Fifty (50) mines shall be placed in the container and submerged in desensitizer which is maintained at the degassing or storage temperature. The recommended atmospheric exposure between degassing or storage and test is five (5) seconds. Rebalance the beam balance and record the second reading. The difference

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Insert the shorting plug specified on the equipment list into connector number 1 of the monitor and test cable. The monitor lite of the XM67 Test Set should light and the audible alarm should function. Remove the shorting plug and repeat the test until all ten (10) connectors have been checked. Remove the shorting plug (9227399) from the P1 receptacle mounted on the container and replace it with a ten-lite test set (see dwg. 9227398). Again insert the shorting plug mentioned above into connector number 1 of the monitor and test cable. Lite number 1 of the test set connector to the canister in the corresponding numbered bay. Repeat the test until all ten (10) connectors have been checked. Each connector shall be immediately mated to the canister in the numbered bay corresponding to the lighted lite on the ten-lite test set prior to checking the next connector. After all ten (10) canisters have been mated to the monitor and test cable, remove the ten-lite test set and replace it with the shorting plug (9227399). Depress the ready button on the XM67 Test Set and insure that the green ready lite lights. If the XM67 Test Set does not operate as stated above or if any lite on the ten-lite test set does not light when its corresponding numbered connector is shorted, the loaded container shall be rejected. This is a non-destructive test; containers so tested may be returned to the lot.

4.4.12 Proving ground test.-These test shall be conducted as stated in the applicable test program requests indicated by the design agency.

4.4.13 Strapping machine verification.-The test sample shall be assembled to a fixture as specified on the equipment list and tested for strap tension.

4.4.14 Leakage test.-The canister with the indicator eye assembly and EOD tube assembly installed and prior to loading shall be placed into a fixture as specified on the equipment list and tested for leakage. This is a non-destructive test, canisters that meet the requirement may be returned to the lot.

4.4.15 Submersion.-The assembly shall be placed upright in a container filled with at least 4 inches of desensitizer (TCTFE), and at the specified time the height of exposure shall be measured. Any assembly which fails to comply with the requirement specified in 3.13 shall be classed defective. This is a non-destructive test.

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accordance with 6.3.5 for all the equipment coded as "Contractor Design" in the "Part Number" column. These contractor designs must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6.

6.3.3 Equipment list codes.-The inspection equipment as defined in 6.3.3 and 6.3.4 will be designated in the EL by the following codes:

CD - Contractor design on controlled contractor format and/or commercial equipment.
MU - Army design, mandatory for use.

6.3.4 Army designs.-Army designs are reflected on detailed drawings which completely depict all the information necessary for the fabrication of the item of inspection equipment. The contractor need provide no design when an Army design is listed for an item of inspection equipment. Army design fall into two basic classifications: Mandatory (designated "MU") and non-mandatory (indicated by drawing or Federal Stock Number). When an equipment list references mandatory Army Designs, the contractor shall comply with, and use these designs accordingly. The contractor may, however, in connection with non-mandatory designs, design alternate inspection equipment or use comparable commercial equipment to facilitate his operations. Such contractor-prepared designs (see 6.3.5) or commercial equipment selections must be approved by the Government prior to fabrication or procuring of the equipment. Designs shall be submitted for approval as specified in 6.3.6.

6.3.5 Contractor designs.-Contractor designs are designs of inspection equipment for which the Government has assigned design responsibility to the contractor. Contractor designs shall be supported by detailed drawings which depict all information necessary to completely fabricate, calibrate and operate an item of inspection equipment. This requires that the necessary views, dimensions, materials, finishes, notes and operating and calibration instructions be properly depicted in accordance with approved practices to the extent that further calculation or clarification will not be required. Unless otherwise specified, contractor designs may be developed on the format the contractor normally employs in his equipment design procedure provided such format reflects the detail and information specified above, subject to the following controls: All submitted contractor designs shall conform to Specification MIL-D-1000, Category E, Form 2. Legibility and reproducibility shall permit conventional making of clearly understandable, high-contrast reproductions. Contractors shall

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a. Canister lot No. and manufacturer.
b. Type of test and equipment utilized.
c. Results of environmental conditioning.
d. Results of functioning, including ejection, canister breakup, dispersion pattern and number of mines emplaced.

3.9.1 Static functioning.-The loaded canister assembly when functioned after being conditioned to 120°F shall break apart without causing damage to the mines nor shall any mine detonate upon impact. The canister shall be assembled less retention base assembly, retention ring assembly, monitor clip assembly and electronic opener.

3.9.2 Vibration.-The loaded and completely assembled canister when conditioned to 120°F and vibrated shall not detonate, lose any desensitizer (TCTFE) or suffer any permanent damage. Upon completion of this test, static functioning of the canister shall be conducted as required in 3.9.1.

3.9.3 Flight test.-The loaded canister assembly when assembled to its dispenser shall release freely during flight without damage to any of its components, after which the canister shall break open in the time frame specified, spilling its contents. There shall be no damage to or detonation of the mines during the breaking of the canister or upon impact with the ground.

3.10 Tension verification of strapping machine.-Strap tension shall be 140 pounds, min., when tested as specified in 4.4.13.

3.11 Workmanship.-All parts and assemblies shall be fabricated, loaded and assembled in a thorough, workmanlike manner. They shall be free of burrs, sharp edges, cracks, dirt, grease, rust and other foreign matter. The cleaning method used shall not be injurious to any part, nor shall the parts be contaminated by the cleaning agents. Exterior surface coatings shall be continuous except for a few light scratches not exposing base material. All required marking and stamping shall be neat and sharply defined.

3.12 First article testing.-This specification makes provisions for first article testing. Submission of first article quantity by the contractor shall be as specified in the contract.

3.13 Submersion (filler and plug assembled, dwgs. 9234030 and 9234031).-The filler and plug assembly shall submerge to the extent that not more than 1/4 inch of the assembly remains

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4.2 First article inspection

4.2.1 Submission.-The contractor shall submit a sample of forty (40) loaded canister assemblies to a Government proving ground designated by the Contracting Officer for testing in accordance with the provisions of 4.2.2 (see 6.1). All parts and materials shall be the same as will be used in production and shall be obtained from the same sources of supply. Prior to submission, the sample shall be inspected by the contractor to the degree necessary to assure that it conforms to the requirements of the contract.

4.2.2 Inspections to be performed.-Assemblies will be subjected by the Government to the following tests:

4.2.2.1 Desensitizer leakage.-The forty (40) loaded canisters shall be subjected to this test. The test shall be performed as specified in 4.4.8.

4.2.2.2 X-ray of loaded canister assembly.-The forty (40) loaded canisters shall be subjected to this test. The test shall be performed as specified in 4.4.9.

4.2.2.3 Proving ground tests.-After completion of tests specified in 4.2.2.1 and 4.2.2.2, the loaded canisters of the quantity indicated below shall be subjected to these tests. The tests shall be performed in accordance with the test paragraph specified below:

Test	Quantity	Test Procedure Para.
Static Functioning	15	4.4.12
Vibration	15	4.4.12
Flight	10	4.4.12

4.2.3 Rejection.-If any assembly or component fails to comply with any of the applicable requirements, the first article quantity shall be rejected. The Government reserves the right to terminate its inspection upon any failure of an assembly or component in the sample to comply with any of the stated requirements. In the event of rejection, the Government reserves the right to require the contractor to take corrective action and submit a new first article quantity or portion thereof. Until a first article quantity is accepted, the contractor is in no way authorized by the Government to resume regular production unless directed by the Contracting Officer.

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with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable subparagraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.3.2.1 CDU-10/B, prior to loading (See dwg. 9228100)

Categories	Defects	Method of Inspection	Defect Code (See 6.2)
Critical:			
1.	EOD tube damaged, missing or improperly assembled	Visual	01001
2.	Location of hole for indicator eye	Gage	01002
Major:	AQL 100 Percent		
101.	Protective sleeve missing or improperly assembled	Visual/Manual	01003
102.	Subpack flap secured by tape or wire	Visual	01005
103.	Hinge pin does not rotate 360°	Manual	01006
Major:	AQL 0.65 Percent		
104.	Depth of counterbore for indicator eye	Gage	01007
105.	Flushness of the tube assembly	Gage	01008
106.	Protrusion of indicator eye, max.	Gage	01009
107.	Set screw (2) not below flush or missing	Visual	01010
Minor:	AQL 0.65 Percent		
201.	Evidence of poor workmanship (see 3.11)	Visual	01011

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104.	Incorrect number of opener plate spacers (when applicable - see dwg. 9235641)	Visual	03006
105.	Gap between opener mounting plate and support adapter	Gage	03007
106.	Gap between opener switch mounting plate and opener mounting plate	Gage	03008
107.	Improper routing of electrical special purpose cable assembly and electronic opener cable (see dwg. 9228100)	Visual	03009
108.	Staking of retention rods and stamped nuts missing or improper	Visual	03010
109.	Explosive opener cable assembly pinched between base plate and opener plate or between opener plate and electronic opener	Visual/Manual	03011
110.	Loose, cut or exposed wire or lanyards (after switch assembly, before and after cover assembly)	Visual	03012
111.	Position and seating of explosive opener assembly	Gage/Manual	03028
112.	Explosive opener not tightly seated	Manual	03014
113.	Lead wires tied to lanyard (just prior to assembly of lid assembly)	Visual	03015
125.	Distance to EOD marking from top of canister, max.	Gage	03001
126.	Improper positioning of lead wires and ground wire of electronic opener (just prior to installation of lid assembly) (see dwg. 9235641)	Visual	03029

Major:	AQL 1.00 Percent		
114.	Torque of retention rod nuts	Test	03016
115.	Torque of electronic opener and opener switch assembly mounting screws	Test	03017
116.	Electronic opener cable and explosive opener cable improperly mated and clamped	Visual	03018
117.	Any component missing or damaged	Visual	03019

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202. Improper position of special purpose electrical cable assembly (see sheet 2 of dwg. 9228100)	Visual	04007
203. Evidence of poor workmanship (see 3.11)	Visual	04008

4.3.2.5 Container for CDU (see dwg. 9234891)

Categories	Defects	Method of Inspection	Defect Code
Critical:			
1.	Indicator eye cannot be read or indicates a low desensitizer level	Visual	05001
Major:	AQL 0.65 Percent		
101.	Torque of screw(s) securing cover	Test	05002
102.	Marking incorrect, missing, misleading or unidentifiable	Visual	05003
103.	Safety panel(s) missing or damaged	Visual	05004
Minor:	AQL 1.00 Percent		
201.	Metallic seal(s) missing	Visual	05005
202.	Evidence of poor workmanship (see 3.11)	Visual	05006

4.3.3 Testing

4.3.3.1 Electrical continuity of liquid elvel sensor (see 3.2.2), Critical Defect, Defect Code 06001.-Each liquid level sensor shall be subjected to this test prior to loading. If any sensor fails to comply with the applicable requirement, it shall not be used in assembly of the canister. The test shall be performed as specified in 4.4.1.

4.3.3.2 Shunt verification of explosive opener fixed delay assembly, prior to loading (see 3.2.3), Critical Defect, Defect Code 07001.-This test shall be performed 100 percent. Any explosive opener fixed delay assembly which fails to comply with the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.2, using equipment in accordance with 4.3.4.

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4.3.3.9 X-ray of loaded canister assembly (see 3.5).- Each loaded canister shall be x-rayed as specified in 4.4.9, using equipment in accordance with 4.3.4.

4.3.3.9.1 (TCTFE) level, Major Defect, Defect Code 12002.- Any loaded canister assembly which fails to comply with the requirements as specified in 3.5.1 shall be classed defective and removed from the lot.

4.3.3.9.2 Mines in or above filler and plug area, Critical Defect, Defect Code 12003.-Any loaded canister assembly which fails to comply with the requirements as specified in 3.5.2 shall be classed defective and removed from the lot.

4.3.3.9.3 Subpack positioning, Major Defect, Defect Code 12004.-Any loaded canister assembly which fails to comply with the requirements as specified in 3.5.3 shall be classed defective and removed from the lot.

4.3.3.10 Continuity of the electrical circuits of the loaded canister (see 3.6), Major Defect, Defect Code 13001.-This test shall be performed 100 percent after assembly and retorquing of the retention rod nuts. Any canister which fails to comply with the specified requirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.10, using equipment in accordance with 4.3.4.

4.3.3.11 Electrical continuity of loaded container (see 3.8), Critical Defect, Defect Code 14001.-this test shall be performed 100 percent. Any loaded container which fails to comply with the specified reqirement shall be classed defective and removed from the lot. The test shall be performed as specified in 4.4.11, using equipment in accordance with 4.3.4.

4.3.3.12 Proving ground tests.-The proving ground tests as called for below shall be for informational purposes only. In the event that a month's production contains canisters from more than one (1) lot, the samples selected shall be representative of each lot contained in that month's production.

4.3.3.12.1 Vibration.-A sample of four (4) loaded canisters shall be randomly selected from each month's production and shipped to a Government-owned proving ground as designated by the Contracting Officer. The test shall be conducted as specified in 4.4.12. Upon completion of this test, the static functioning test shall be conducted.

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evaluation, production and receiving inspection, overhaul, shipping, storage, identification of stock, ordering and storage of replacement parts, inspection of items at overhaul, general maintenance of equipment, construction, survey, and whenever inspection equipment drawings are needed.

4.3.4.2 Supply and maintenance.-Supply and maintenance of the equipment listed on the EL's shall be in accordance with MIL-I-45607.

4.4 Test methods and procedures

4.4.1 Electrical continuity of liquid level sensor, prior to loading.-The liquid level sensor shall be tested using fixtures, instruments, and the method as shown on the equipment list and (dwg. 9202901). This is a non-destructive test; sensors that meet the requirements of 3.2.2 may be returned to the lot.

4.4.2 Shunt verification of explosive opener fixed delay assembly, prior to loading.-The equipment specified on the equipment lists shall be assembled to the explosive opener lead assembly and the opener assembly checked for compliance with the requirement specified in 3.2.3. A maximum current of one (1) milliampere shall be applied to the explosive opener. This is a non-destructive test; parts so tested may be returned to the lot.

WARNING: This test shall be performed behind a suitable barricade.

4.4.3 Electrical continuity, resistance and shorts of cable assembly, prior to loading.-The cable assembly shall be assembled to the equipment specified on the equipment list and checked for compliance with 3.2.4. This is a non-destructive test; parts so tested may be returned to the lot.

4.4.4 Functioning of electronic opener, prior to loading .-The electronic opener shall be assembled to a fixture as shown on the equipment list and tested for compliance with 3.2.5. This is a non-destructive test; parts so tested may be returned to the lot.

4.4.5 Continuity and resistance of the opener switch assembly, prior to loading.-The opener switch assembly shall be assembled to a fixture as shown on the equipment list and tested for compliance with 3.2.6. This is a non-destructive test; parts so tested may be returned to the lot.

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between the two readings is the total buoyant force for fifty (50) mines and shall be less than the specified requirements. Any sample which fails to staisfy the specified requirement shall be classed defective and the degassing batch or storage container rejected. Records shall be maintained of each test sample identifying whether items were stored or degassed, temperature of desensitizer during storage or before and after degassing. Records shall also be maintained of the temperature of desensitizer used in this test on a minimum basis of at least twice per shift.

4.4.8 Desensitizer (TCTFE) leakage (see 3.4).-The loaded canister shall be placed into a fixture as specified on the equipment list and tested for desensitizer leakage. This is a nondestructive test; canisters so tested may be returned to the lot.

4.4.9 X-ray of loaded canister assembly.-The loaded canister shall be x-rayed in the vertical position one (1) hour minimum after sealing. There shall be no shaking of the canister or mixing of its contents for the one (1) hour period immediately prior to x-raying. The fixture, x-ray equipment, positioning of canister, positioning of x-ray equipment, exposure mode, etc., shall be as specified in the applicable equipment list. Any deviations from established procedure, as specified in the equipment list, shall be sent to Picatinny Arsenal, ATTN: SMUPA-ND8 (see 6.4). The developed films identified as to the canister lot and canister serial number shall be retained for a minimum period of one (1) year. Radiographic equipment, operators and procedures shall be qualified in accordance with MIL-R-11470. In addition, the sensitivity of radiographic density of the film used shall be at least 1.25 percent and the radiographic density of the film used shall be of a quality which will produce a definite internal structure as well as showing the level of desensitizer. This is a non-destructive test; canisters so tested shall be returned to the lot, if acceptable.

4.4.10 Continuity of the electrical circuit of the loaded canister (see 3.6).-The equipment specified on the equipment list shall be assembled to the loaded canister and the canister checked for compliance with the specified requirement. This test shall be performed after assembly and re-torquing of the retention rod nuts.

4.4.11 Electrical continuity of the loaded container.-Prior to placing the cover on the container and connecting the monitor and test cable to the canisters, connect an XM67 Test Set (see dwg. 9211625) to the P2 receptacle mounted on the container.

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5. PREPARATION FOR DELIVERY

5.1 Packing.-The loaded canister shall be packed in accordance with dwg. 9234891.

5.2 Marking.-Marking shall be in accordance with dwg. 9234891 and Interstate Commerce Commission Regulations 49 CFR 171-179.

5.3 Shipping.-When shipments from more than one lot are shipped as a carload, each lot shall be kept separate, and the division between lots clearly indicated to prevent mixing of the lots in transit.

6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167.
c. Provisions for submission of First Article Samples.

6.2 Inspection code numbers.-The five-digit code numbers assigned to the inspections herein are to facilitate data collection and analysis by the Government. These code numbers are also used to correlate the characteristics cited on Equipment Lists with the inspections listed in this specification. These code numbers shall be listed on drawings of equipment designs and inspection data submitted by the contractor.

6.3 Inspection equipment.-The contractor shall design and submit for approval inspection equipment as required by the referenced Equipment Tabulation (ET) and each applicable Equipment List (EL) in accordance with the instructions of paragraph 6.3.1 through 6.3.6.

6.3.1 Equipment Tabulation.-This tabulation (SMUPA Form 1011) is an index of all applicable EL's.

6.3.2 Equipment Lists (EL).-This list (SMUPA Form 1010) indicates the availability of inspection equipment designs by showing in the "Part of Identifying No." column, the numbers of drawings or Federal Stock numbers of existing equipment designs, or codes as indicated in paragraph 6.3.3. Design action required of the contractor is described in paragraphs 6.3.4 and 6.3.5. The contractor will be required to prepare detailed drawings in

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submit three (3) copies of final designs as a flat set. Designs shall be submitted for approval as specified in 6.3.6.

6.3.6 Submission of contractor designs.-All submitted designs shall contain a reference to the applicable 5-digit code number contained in Section 4 of this specification and the appropriate component or assembly drawing number and revision letter to which the specific design applies. Unless otherwise specified on the EL, all designs of equipment for inspection of defects classified as critical and major shall be submitted for approval to the Commanding Officer, Picatinny Arsenal, ATTN: SMUPA-ND8. All other designs of inspection equipment shall be approved by the inspection element of the agency administering the contract. Submission shall be as directed to the contracting officer. Partial submission of inspection equipment designs is permissible and encouraged. However, the Arsenal completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will be accomplished normally, within one month after receipt.

6.4 Submission of test results.-Information as required in Section 3 of this specification shall be forwarded to:

Commanding Officer
Picatinny Arsenal
ATTN: SMUPA-ND8
Dover, New Jersey 07801

Custodian:
Army-MU

Preparing activity:
Army-MU

Project Number: 1345-A-267

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